

Human Review: bidual stability of uniform property (S)

Original automatic proof: `solution_packet.pdf`

Checked date: 10 June 2026

Status: Manually checked.

Verdict: Correct.

Human Comments

The solution is correct. Once the idea is mentioned, the argument is quite easy to spot: choose $X = L_1[0, 1]$, so that the underlying measure is atomless and the positive results of the paper apply to X . Thus X has uniform property (S).

The obstruction appears only after passing to the bidual. Since

$$X^{**} = (L_\infty[0, 1])^*,$$

one can represent $L_\infty[0, 1]$ as $C(K)$ for its Stone space K , and therefore

$$X^{**} \simeq C(K)^* = M(K).$$

Equivalently, by the representation theory of abstract AL-spaces, this space may be viewed as an $L_1(\nu)$ -space for a measure ν with atoms. In the Stone representation, the atoms are visible as Dirac measures.

The key point is that once the bidual contains at least two atoms, uniform property (S) cannot hold. The packet exhibits two Dirac measures δ_s, δ_t and then chooses

$$u = \frac{3}{4}\delta_s + \frac{1}{4}\delta_t, \quad v = \frac{1}{4}\delta_s + \frac{3}{4}\delta_t.$$

For sufficiently small perturbations, the two atomic coefficients stay positive, and the total variation norms of $u + z$ and $v + z$ remain equal. This gives

$$U_{X^{**}}(u, v; 1/8) = 0 \quad \text{while} \quad \|u - v\| = 1,$$

so X^{**} fails uniform property (S).

Short Version

The example is $X = L_1[0, 1]$. The space X has uniform property (S) because the measure is atomless. But X^{**} can be represented as a measure space with atoms, and the presence of two atoms gives a direct two-coordinate obstruction to uniform property (S).

Further Question

The proof uses two atoms. It would be interesting to understand whether the paper's positive results, or some variant of them, extend to the case where the representing measure has exactly one atom. The present packet does not settle that one-atom case; it only shows that two atoms are already enough to destroy uniform property (S).

Author Follow-Up

After reaching out to the authors, they indicated that they had already come up with the same example for a new version that has not yet been published. This should therefore be recorded as a correct rediscovery/check rather than as a novel contribution.

Review Outcome

Archive this packet as a correct negative answer to bidual stability of uniform property (S), with the side note that the one-atom case remains a natural follow-up question. Also record that the same example was already known to the authors for a forthcoming version, so this should not be counted as an independent novel contribution.